Unified superconductivity from the TGP substrate

Cuprates, phonon-mediated metals, pressure-driven orbital switching, and lanthanide hydrides from three universal constants

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Abstract

Building on the *Theory of Generated Space* (TGP) introduced in [1], we apply the substrate-field framework to superconductivity. The superconducting critical temperature of a material is determined by a single substrate ansatz $T_c = C_0\, A_{\rm orb}^2\, k_d(z)\, M(a)\, \Lambda_E$ combined with four physical blockers (pair-breaking, orbital localisation, spin-fluctuation damping, magnetic moment scattering). We introduce three universal TGP constants — $\beta = 2.527$ (magnetic blocking), $\kappa_{\rm TGP} = 2.012$ (spin-fluctuation coupling calibrated on five paramagnetic d-metals), $\alpha_{\rm PB} = 0.2887\,\mu_B^{-2}$ (Abrikosov–Gorkov lanthanide pair-breaking) — plus tabulated atomic inputs $\{N(E_F), I, \mu_{\rm eff}\}$. The framework reproduces $T_c^{\rm obs}$ across thirty-one materials in five classes with $r(\log T_c) = 0.875$, $\mathrm{RMS}_{\log} = 0.346$, without class-specific re-fitting. Critically, the spin-fluctuation coupling $\lambda_{\rm sf}$, previously assigned empirically per material, is replaced by a TGP-structured closed-form expression with a single fitted constant $\kappa_{\rm TGP}$ and tabulated Stoner parameters, improving the fit from $r = 0.874$ to $r = 0.943$ while reducing eleven phenomenological inputs to one. We state five named falsifiable predictions, most notably $T_c(\mathrm{LuH_{10}}, 170\,\mathrm{GPa}) \approx 200$–$250\,\mathrm{K}$ and $T_c(\mathrm{Hg1245}, \mathrm{SrTiO_3}) \approx 178\,\mathrm{K}$, and additionally tabulate TGP predictions for ~ 25 currently investigated superconductor candidates (lanthanide / scandium superhydrides, ambient-pressure $\mathrm{Mg_2XH_6}$ and $\mathrm{Li_2XH_6}$ families, bilayer nickelates, kagome Cr-based compounds, nitride-hydrides) including a "$4f^7$ killer" null prediction ($T_c \approx 0$ for $\mathrm{GdH_9}$, $\mathrm{TbH_9}$, …, $\mathrm{TmH_9}$) that directly tests Abrikosov–Gorkov pair-breaking in the substrate picture. We flag two outstanding open subproblems on the same footing as the positive results: the Ce/Kondo case (mixed-valence screening of $\mu_{\rm eff}$ required to match $T_c \sim 100\,\mathrm{K}$ for $\mathrm{CeH_9}$, labelled P7.3), and the partial $n^{0.7}$ correction to the cuprate $\sqrt{n}$ layer factor (L1 in Sec. 10).

Keywords: superconductivity, TGP substrate, cuprates, phonon coupling, hydrides, lanthanide pair-breaking, Abrikosov–Gorkov.

Contents

1 Introduction **3**

2 Substrate ansatz for T_c **3**

3 Cuprate class (P6.A) **4**

4 Phonon-mediated class (P6.B) **4**

5 Orbital switching under pressure (P6.C) **4**

6 Magnetic blocking and first-principles λ_{sf} (P6.D + P7.1) **5**
6.1 TGP-derived λ_{sf} formula (calibrated on five d-metals) 5
6.2 Atomic inputs 5
6.3 Consistency check: does derived λ_{sf} predict T_c better? 6

7 Lanthanide pair-breaking (P7.2) **6**

8 Master validation **7**

9 Falsifiable predictions **7**
9.1 Testability hierarchy of the five named predictions 8
9.2 Open-candidate survey (2025–2026) 8

10 Discussion and limitations **10**

11 Conclusion **11**

1 Introduction

Superconductivity remains a canonical test of any theory of emergent physics. Within standard frameworks, cuprates, iron pnictides, phonon mediated metals, pressure-driven exotica and lanthanide hydrides are described by distinct and often disconnected mechanisms — BCS for weak-coupling phonons, Eliashberg for strong-coupling, spin-fluctuation mediated pairing for high-T_c cuprates, and magnetic moment pair-breaking for lanthanide-containing hydrides.

Building on the TGP framework [1], we demonstrate that a single substrate ansatz for the pairing scale, corrected by four physical blockers, predicts $T_c^{\rm obs}$ across all five classes. The key physical idea is that the Cooper pair wavefunction feeds into the TGP substrate through its orbital amplitude $A_{\rm orb}$; the substrate responds at a characteristic length $a^*_{\rm TGP} = 7.725\,a_{\rm B} \approx 4.088\,\text{Å}$; and pair formation is blocked by several system-specific effects whose magnitudes we quantify.

Structure. Section 2 introduces the core substrate ansatz. Sections 3–7 develop the six sector modules (cuprates; phonon-mediated; orbital switching; magnetic blocking; TGP-derived $\lambda_{\rm sf}$; lanthanide pair-breaking). Section 8 presents the master validation on 31 materials. Section 9 states the five falsifiable predictions. Section 10 discusses limitations, including two open subproblems that are stated here for full transparency: (i) the Ce/CeH_9 Kondo-screened pair-breaker, where the naive Hund moment $\mu_{\rm eff} = 2.54$ over-predicts the observed T_c by a factor $\sim$3 and requires a mixed-valence correction (P7.3); and (ii) the cuprate layer-factor approximation $\sqrt{n}$ (L1).

2 Substrate ansatz for T_c

Following the TGP core [1], the substrate possesses a universal spatial harmonic

$$a^*_{\rm TGP} = 7.725\,a_{\rm B} \approx 4.088\,\text{Å}, \tag{1}$$

and couples to matter through the orbital amplitudes

$$A_s = -0.111, \quad A_{sp} = 0.207, \quad A_d = 0.310, \quad A_f = 2.034, \tag{2}$$

which quantify the overlap between a given electronic orbital character and the substrate soliton profile.

Definition 1 (Core substrate ansatz for T_c)**.** *The superconducting transition temperature of a material with in-plane lattice constant a, coordination number z, dominant orbital character* orb $\in \{s, sp, d, f\}$ *and substrate pairing scale Λ_E is*

$$\boxed{T_c^{\rm core} = C_0\, A_{\rm orb}^2\, k_d(z)\, M(a)\, \Lambda_E/k_B,} \tag{3}$$

with $C_0 = 48.8222$ a universal TGP constant, $k_d(z)$ a Monte-Carlo-computed coordination factor (Table 1), and $M(a) = \exp\!\big[-\frac{1}{2}(a - a^*_{\rm TGP})^2/\sigma^2\big]$ the substrate-resonance factor ($\sigma = 0.25$ Å).

Table 1: TGP coordination factor $k_d(z)$ from Monte-Carlo of the substrate soliton on a local environment with coordination z.

z	$k_d(z)$	typical geometry	examples
4	0.893	tetrahedral	diamond, sp^3
6	2.202	octahedral	rock-salt
8	2.936	BCC	V, Nb, W
12	4.403	FCC/HCP	Al, Cu, Pd

Eq. (3) is the *bare* TGP prediction. Physical blockers reduce it:

$$T_c \;=\; T_c^{\mathrm{core}} \cdot B_{\mathrm{orb}}(P) \cdot B_{\mathrm{mag}}(\lambda_{\mathrm{sf}}) \cdot B_{\mathrm{PB}}(\mu_{\mathrm{eff}}), \tag{4}$$

with the three factors detailed in Secs. 5, 6 and 7 respectively. The scale Λ_E depends on the pairing class and is developed below.

3 Cuprate class (P6.A)

In cuprates the pairing is of d-wave character and involves the Zhang–Rice singlet (Cu $3d$ + 2× O $2p$). The TGP model reads

$$T_c^{\mathrm{cup}} = K_{\mathrm{dw}}\, k_d(z_{\parallel})\, C_0\, A_{\mathrm{ZR}}^2\, M(a_{\parallel})\, \sqrt{n}\, \Lambda_E^{\mathrm{cup}}/k_B, \tag{5}$$

with: $K_{\mathrm{dw}} = 3.498$ (d-wave node boost × van Hove enhancement × ZR correction); $A_{\mathrm{ZR}}^2 = A_d^2 + 2A_p^2 = 0.181$; $z_{\parallel} = 8$ (planar coordination), $k_d = 2.936$; layer factor $\sqrt{n}$ (saturated for $n > 3$); $\Lambda_E^{\mathrm{cup}} = 0.0513\,\mathrm{meV}$ (2.6× smaller than BCS 0.131 meV – electron-mediated pairing, not phonon).

Fit on 8 cuprates. $r(\log T_c) = 0.96$, $\mathrm{RMS}_{\log} = 0.19$. Residuals are systematic: the $\sqrt{n}$ factor *undershoots* $n = 2, 3$ records by -0.10 to -0.17 in $\log_{10}$, and overshoots $n = 1$ materials by $+0.15$ to $+0.35$. This motivates an open question — whether the correct layer factor is $n^{0.7}$ with an offset rather than $\sqrt{n}$ — which we leave to future work.

Interface factor. Coupling a cuprate to a Fuchs–Kliewer interface (e.g. $SrTiO_3$) enhances Λ_E^{cup} by the substrate boost factor, raising T_c toward room-temperature regimes (Sec. 9).

4 Phonon-mediated class (P6.B)

For materials with phonon-mediated pairing (Al, Pb, Nb, V, H_3S, LaH_{10}, MgB_2, Nb_3Sn, ...) the substrate pairing scale depends on the Debye frequency ω_{ph} as

$$\Lambda_E^{\mathrm{eff}}(\omega_{\mathrm{ph}}) = \Lambda_0 \Big(\frac{\omega_{\mathrm{ph}}}{\omega_0}\Big)^{\alpha_B}, \tag{6}$$

with $\alpha_B = 1.04$, $\Lambda_0 = 0.0962\,\mathrm{meV}$, $\omega_0 = 15\,\mathrm{meV}$ (Al reference). The exponent $\alpha_B \simeq 1$ is physically interpretable: the substrate amplitude Φ oscillates at ω_{ph} and transfers energy proportional to it, to leading order. Eq. (3) with (6) yields

$$T_c^{\mathrm{ph}} = C_0\, A_{\mathrm{orb}}^2\, k_d(z)\, M(a)\, \Lambda_0 \Big(\tfrac{\omega_{\mathrm{ph}}}{\omega_0}\Big)^{1.04}/k_B. \tag{7}$$

Calibration → validation. Fitted on 9 hydrides (α_B, Λ_0 free), it validates $r(\log T_c) = 0.877$, $\mathrm{RMS}_{\log} = 0.32$ on a mixed set of 16 materials, including: MgB_2 predicted 31.4 K vs observed 39 K (ratio 0.80); $FeSe/SrTiO_3$ predicted 86 K vs observed 65 K (1.33); LaH_{10} predicted 368 K vs observed 250 K (1.47, overshoot within scale).

5 Orbital switching under pressure (P6.C)

Elements with partially localised d or f shells (Ce, Yb, lanthanides) exhibit a pressure-induced delocalisation. In TGP this is captured by a pressure-dependent amplitude mixing:

$$A_{\mathrm{eff}}(P) = \eta(P)\, A_d, \qquad \eta(P) = \eta_0 + (1-\eta_0)\big[1 - e^{-P/P_{\mathrm{scale}}}\big], \tag{8}$$

with $\eta_0 \in [0.1, 0.5]$ (ambient localisation) and P_{scale} a material-specific pressure scale. The orbital blocker is $B_{\mathrm{orb}}(P) = \eta(P)^2$.

Empirical anchors. Ce ($4f^1$) at 5 GPa yields $P_{\rm scale}^{\rm Ce} = 5.8\,$GPa (fitted to Ce superconductivity onset). Yb ($4f^{14}$) was initially assumed $P_{\rm scale} \sim 10\,$GPa, but this was falsified by the Yb_4H_{23} measurement $T_c = 11.5\,$K at 180 GPa [7], implying $P_{\rm scale}^{\rm Yb} = 552\,$GPa. The revised value lowers the YbH_9 prediction from 215 K to 38 K. A linear extrapolation across $4f^n$ gives $P_{\rm scale}(4f^n) \simeq 5.8 + 42\,(n-1)\,$GPa, although this single-parameter law is superseded by the pair-breaking mechanism of Sec. 7 for the heavier lanthanides.

6 Magnetic blocking and first-principles $\lambda_{\rm sf}$ (P6.D + P7.1)

Paramagnons suppress s-wave-like pairing by spin-flip scattering of Cooper partners. The TGP magnetic blocker is

$$B_{\rm mag}(\lambda_{\rm sf}) = \frac{1}{1+\beta\,\lambda_{\rm sf}}, \qquad \beta = 2.527, \tag{9}$$

with β a universal TGP constant fitted across the non-cuprate dataset. $\lambda_{\rm sf}$ is the spin-fluctuation coupling. Prior to the present work, $\lambda_{\rm sf}$ was assigned *empirically* per material (eleven free numbers across V, Nb, Ta, Mo, Pd, Fe, Co, Ni, Cr, Rh, Ru). We replace this per-material tuning by a TGP-motivated closed-form expression calibrated on five paramagnetic d-metals.

6.1 TGP-derived $\lambda_{\rm sf}$ formula (calibrated on five d-metals)

In the Eliashberg-type analysis, the paramagnon is a local fluctuation of the substrate Φ. Coupling to a d-band electron through the substrate amplitude $A_{\rm eff} = \eta\,A_d$, and integrating over the paramagnon spectral density with a Stoner regulator, yields the closed-form expression

$$\boxed{\lambda_{\rm sf}^{\rm TGP} = \kappa_{\rm TGP}\,A_{\rm eff}^2\,k_d(z)\,N(E_F)\,g(I{\cdot}N(E_F)), \qquad g(x) = \frac{x^2}{\sqrt{1+0.25\,x^4}},} \tag{10}$$

with $N(E_F)$ the density of states at the Fermi level (units st/eV/atom), I the Stoner parameter, and $\kappa_{\rm TGP} = 2.012$ a single universal coupling calibrated on five paramagnetic d-metals ({V,Nb,Ta,Mo,Pd}). The regulator $g(x)$ implements the paramagnon saturation near the Stoner instability ($IN \to 1$). We emphasise that this is a *TGP-structured* formula with one fitted constant ($\kappa_{\rm TGP}$), not an *ab initio* derivation from electronic structure: the analytic dependence on $\{A_d, k_d, N(E_F), I\cdot N(E_F)\}$ is the TGP content, while the overall normalisation is set empirically. The validation criterion is whether the same formula, once calibrated, transfers without retuning to an independent test set (Fe-pnictides / chalcogenides); it does, at the 0.9–1.0× level (Table 2).

6.2 Atomic inputs

The atomic inputs $N(E_F)$ and I are tabulated from Janak 1977 [3] and Moruzzi–Janak–Williams 1978 [4].

Table 2: Fit and validation of $\lambda_{\rm sf}^{\rm TGP}$ using Eq. (10). Fit: V, Nb, Ta, Mo, Pd ($\kappa_{\rm TGP} = 2.012$). Validation: Fe-pnictides/chalcogenides.

Material	$N(E_F)$	I	IN	$\lambda_{\rm sf}^{\rm obs}$	$\lambda_{\rm sf}^{\rm pred}$	status
V	1.35	0.72	0.97	0.60	0.70	fit
Nb	1.24	0.57	0.71	0.20	0.34	fit
Pd	1.46	0.59	0.86	0.80	0.88	fit
FeSe bulk	—	—	0.80	0.90	0.69	validation
Ba122-Co	—	—	0.54	0.30	0.29	validation (0.98×)
LaFeAsO	—	—	0.66	0.50	0.47	validation (0.93×)
NdFeAsO-F	—	—	0.54	0.30	0.29	validation (0.98×)

6.3 Consistency check: does derived $\lambda_{\rm sf}$ predict T_c better?

Replacing the eleven empirical $\lambda_{\rm sf}$ values with the TGP-derived ones in Eq. (4) *improves* the Pearson correlation on the combined T_c dataset,

$$r_{\rm empirical} = 0.874 \longrightarrow r_{\rm derived} = 0.943. \tag{11}$$

The parameter count is reduced from eleven phenomenological numbers to one universal constant ($\kappa_{\rm TGP}$) plus two atomic inputs. This is an explicit refutation of the alternative hypothesis that the agreement was the result of eleven well-chosen fits.

Scope of validity. The formula (10) applies in the paramagnetic regime $IN < 0.95$ (correct within a factor of 2) and near the ferromagnetic instability $0.95 \le IN < 1.5$ (regulator saves Fe). For ordered ferromagnets with $IN \ge 1.5$ (Co, Ni) the formula over-predicts by 2–3×, although all such cases are already magnetically suppressed to $T_c \to 0$. Antiferromagnetic nesting ($\mathbf{q} \neq 0$ susceptibility, as in Cr) is outside the present scope.

7 Lanthanide pair-breaking (P7.2)

The naive prediction for lanthanide superhydrides LnH_9 at $\eta = 1$ (full delocalisation under pressure) gives $T_c \approx 200\,$K for all Ln. Experiment contradicts this sharply: PrH_9 has $T_c \approx 5\,$K [5] and NdH_9 has $T_c \approx 4.5\,$K [6]. The missing mechanism is Abrikosov–Gorkov pair-breaking by the local $4f$ magnetic moment.

Definition 2 (TGP lanthanide pair-breaking factor)**.** *The pair-breaking blocker is*

$$\boxed{B_{\rm PB}(\mu_{\rm eff}) = \exp\!\big(-\alpha_{\rm PB}\,\mu_{\rm eff}^2\big),} \tag{12}$$

with $\mu_{\rm eff} = g_J\sqrt{J(J+1)}$ *the Hund ground-state moment of the* Ln^{3+} *ion and* $\alpha_{\rm PB}$ *a universal TGP constant.*

Physical basis. Local $4f$ moments disrupt the Φ-substrate phase through spin-flip scattering. In the Abrikosov–Gorkov formalism $\ln(T_c^{(0)}/T_c) = \psi(\frac{1}{2}+\Gamma_{\rm sf}/2\pi T_c) - \psi(\frac{1}{2})$; in the regime $\Gamma_{\rm sf} \gg T_c$ this reduces to an exponential suppression with $\Gamma_{\rm sf} \propto \mu_{\rm eff}^2$, recovering Eq. (12).

Fit. $\alpha_{\rm PB} = 0.2887\,\mu_B^{-2}$, fitted on PrH_9 ($\mu_{\rm eff} = 3.58$, $T_c = 5\,$K) and NdH_9 ($\mu_{\rm eff} = 3.62$, $T_c = 4.5\,$K) with a base temperature $T_c^{\rm base} = 143\,$K.

Validation. $\mathrm{RMS}_{\log} = 0.316$ across five known LnH_9 materials (LaH_{10}, CeH_9, PrH_9, NdH_9, YH_9). The outlier is Ce ($T_c^{\rm pred} = 22\,$K vs $T_c^{\rm obs} = 100\,$K), where Kondo screening reduces $\mu_{\rm eff}$ by a factor ~ 2; this is an open problem (P7.3).

Table 3: TGP predictions for LnH_9 at $P \sim 150\,$GPa ($T_c^{\rm base} \approx 200\,$K). "killer" = heavy $4f$, $T_c \to 0$.

Ln	$4f^n$	$\mu_{\rm eff}/\mu_B$	$B_{\rm PB}$	$T_c^{\rm pred}$ [K]	comment
La	0	0.00	1.0	**200**	empty $4f$, confirmed
Ce	1	2.54	0.155	31 (Kondo: ∼100)	mixed valence
Pr	2	3.58	0.025	5	confirmed
Nd	3	3.62	0.023	5	confirmed
Sm	5	1.55	0.500	**100**	testable
Gd–Tm	7–12	7.6–10.6	$\lesssim 10^{-8}$	≈ 0	killers
Yb	13	0.00	1.0	(38, $\eta < 1$)	η-limited
Lu	**14**	**0.00**	**1.0**	**200–250**	**key prediction**

8 Master validation

Combining all modules into Eq. (4) with Sec. 3–7 yields a parameter-free prediction for each material given $\{a, \mathrm{orb}, z, \omega_{\mathrm{ph}}, P, N(E_F), I, \mu_{\mathrm{eff}}\}$.

Table 4: Parameters used in the master validation, grouped by *provenance*. Category C ("Core"): inherited unchanged from the TGP core paper [1] and not re-fitted here. Category F ("SC-fit"): fitted once on a specific calibration subset (stated in the Notes column) and then held fixed for the entire 31-material master validation. Category D ("SC-derived"): follows from an analytic TGP construction plus atomic inputs and is not a free parameter. No parameter is re-fitted per-material; no parameter is re-fitted per-class.

Parameter	Value	Cat.	Notes / calibration set
C_0	48.8222	C	substrate soliton [1]
a^*_{TGP}	4.088 Å	C	substrate harmonic [1]
$\{A_s, A_{sp}, A_d, A_f\}$	$-0.111, 0.207, 0.310, 2.034$	C	orbital amplitudes [1]
K_{dw}	3.498	F	cuprate d-wave boost; fitted on 8 cuprates (P6.A)
Λ_E^{cup}	0.0513 meV	F	cuprate pairing scale; fitted on 8 cuprates (P6.A)
α_B	1.04	F	phonon exponent; fitted on 9 hydrides (P6.B)
Λ_0, ω_0	0.0962 meV, 15 meV	F	phonon reference; fitted on 9 hydrides (P6.B)
β	2.527	F	magnetic-blocking exponent; fitted on non-cuprate d-metals
κ_{TGP}	2.012	F/D	λ_{sf} coupling, calibrated on {V,Nb,Ta,Mo,Pd}; analytic regulator
α_{PB}	$0.2887\,\mu_B^{-2}$	F	lanthanide pair-breaking; fitted on $\{PrH_9, NdH_9\}$

Scope of each fit. Each F-category constant above is fitted once on a *named* calibration subset (P6.A cuprates; P6.B hydrides; the five paramagnetic d-metals for κ_{TGP}; the two PrH_9/NdH_9 lanthanides for α_{PB}), and then used unchanged on the rest of the 31-material dataset plus the $\sim$25 open candidates of Table 6. No step in the pipeline reads a target T_c^{obs} during the "prediction" phase, and no per-material tuning is applied.

Master plot on 31 materials. Five classes: 9 cuprates (including Hg1223 pressure-quench $T_c = 151$ K [8]); 13 phonon-mediated (Al, Pb, Nb, V, Mo, W, MgB_2, H_3S, LaH_{10}, Nb_3Sn, NbTi, Ba122-Co, ...); 5 Fe-chalcogenides/pnictides (FeSe bulk, FeSe/STO, LaFeAsO, NdFeAsO-F, Ba122-Co); 3 lanthanide hydrides (CeH_9, PrH_9, NdH_9); 1 Yb test case (Yb_4H_{23}).

$$r(\log T_c) \;=\; 0.875, \qquad \mathrm{RMS}_{\log} \;=\; 0.346, \qquad N = 31. \tag{13}$$

Nine of these materials use the first-principles $\lambda_{\mathrm{sf}}^{\mathrm{TGP}}$ of Eq. (10) rather than tabulated empirical values.

9 Falsifiable predictions

Five predictions of the present framework have $O(1)$ leverage on the underlying TGP constants.

Prediction 3 (LuH_{10} at high pressure)**.** *Pure LuH_{10} at $P \simeq 170$ GPa in the $Fm\bar{3}m$ cage structure of LaH_{10} should exhibit*

$$T_c^{\mathrm{pred}}(\mathrm{LuH_{10}}, 170\,\mathrm{GPa}) \;\approx\; 200\text{–}250\,K. \tag{14}$$

Lu^{3+} has $4f^{14}$ closed shell and $\mu_{\mathrm{eff}} = 0$, so $B_{\mathrm{PB}} = 1$. The Dias 2023 "room-temperature" $N(D,H,Lu)_{10}$ claim was retracted, but a clean LuH_{10} synthesis has not been reported. ***Falsification:*** *$T_c < 100\,K$ would require revision of Eq. (12) or of the T_c^{base} estimate for hydrogen-rich cages at $\eta = 1$.*

Prediction 4 (Hg1245 on $SrTiO_3$). *The tier-1 cuprate Hg1245 with $n = 5$ CuO_2 layers, grown by MBE on $SrTiO_3$ with a Fuchs–Kliewer phonon boost, is predicted to satisfy*

$$T_c^{\text{pred}}(\text{Hg1245/SrTiO}_3) \approx 178\,K. \tag{15}$$

This jointly tests the $\sqrt{n}$ layer factor of P6.A and the interface-induced Λ_E^{cup} enhancement. Falsification ($< 100\,K$) would refute the $\sqrt{n}$ scaling.

Caveat on the $\sqrt{n}$ extrapolation. *The $\sqrt{n}$ factor is an empirical approximation fitted on Hgn-series cuprates with $n \leq 3$; residuals on the calibration set suggest a slightly weaker exponent of $\sim n^{0.7}$ (see systematic limitation L1 in Sec. 10). Propagating that uncertainty, the value $178\,K$ in Eq. (15) carries an estimated $\pm 25\,K$ envelope before any interface systematics are included. A measured T_c between 100 and $200\,K$ would be consistent with TGP; only $T_c < 100\,K$ would constitute a* binary *falsification of the combined $\sqrt{n}$ + interface-boost mechanism.*

Prediction 5 (FeSe on $BaTiO_3$). *Interface-enhanced FeSe on a $BaTiO_3$ substrate is predicted to exhibit $T_c \approx 47\,K$, which would falsify the hypothesis that $SrTiO_3$ is uniquely special for Fuchs–Kliewer-boosted SC.*

Prediction 6 (YbH_9 / YbH_{10} at $\geq 300\,\text{GPa}$). *YbH_9 at $300\,\text{GPa}$ is predicted to give $T_c \approx 38\,K$ (not $215\,K$) and YbH_{10} at $400\,\text{GPa}$ gives $\approx 71\,K$, after the Yb_4H_{23}-calibrated $P_{\text{scale}}^{\text{Yb}} = 552\,\text{GPa}$. Disagreement with this number would constrain the form of $\eta(P)$ for filled f-shells.*

Prediction 7 (SmH_9 at $\sim 150\,\text{GPa}$). *Sm^{3+} has $\mu_{\text{eff}} = 1.55$ (anomalously low due to $L - S$ cancellation), giving $B_{\text{PB}} \simeq 0.50$ and hence $T_c \approx 100\,K$. This is an intermediate-moment test of the pair-breaking mechanism (12) — a dedicated probe between the $\mu = 0$ (La, Y, Lu) and $\mu \gg 1$ (heavy Ln) limits.*

9.1 Testability hierarchy of the five named predictions

Not all five predictions above are equally accessible experimentally. Table 5 classifies them by the nature of the required synthesis and the time-horizon on which a definitive measurement is plausible. "Tier-S" = short-term, within current high-pressure diamond-anvil and thin-film capabilities; "Tier-M" = medium-term, requires a new or refined synthesis route; "Tier-H" = high-risk, depends on materials that are theoretically allowed but not yet synthesised.

Table 5: Testability tiers of the five named predictions of Sec. 9. "Falsification leverage" indicates how directly a definitive measurement would bear on a specific TGP equation.

Tier	Prediction	Required capability	Falsification leverage
S	SmH_9 at $\sim$150 GPa	DAC + laser heating	B_{PB} intermediate-moment (Eq. (12))
S	YbH_9 / YbH_{10} at $\geq$300 GPa	DAC high-P	form of $\eta(P)$ for filled f-shell
S	FeSe on $BaTiO_3$	MBE on BTO	specificity of Fuchs–Kliewer boost
M	LuH_{10} at $\sim$170 GPa	clean Lu–H synthesis	T_c^{base} at $\eta = 1$, B_{PB}=1
M	Hg1245/$SrTiO_3$	5-layer cuprate MBE	$\sqrt{n}$ layer scaling + interface Λ_E

Tiers S are the immediate falsification targets. The two Tier-M predictions carry the largest T_c numerical values and are the flagship tests in the long run, but they also have the largest synthesis uncertainty and therefore the weakest "binary" character on a short horizon.

9.2 Open-candidate survey (2025–2026)

Beyond the five named predictions above we apply Eqs. (3)–(12) with *all TGP constants frozen* to materials that are currently under active investigation but for which no T_c has yet been

measured (or for which theory and experiment disagree). We report both the raw TGP value T_c^{raw} and the anchor-normalised value T_c^{anchor}, where the anchor factor rescales the prediction using the nearest calibrated observation in the same class. For high-pressure hydrides the anchor is LaH_{10} at 170 GPa ($T_c^{\mathrm{obs}}/T_c^{\mathrm{raw}} = 250/368 = 0.679$); for nickelates the anchor is Hg1223 applied through the cuprate formula. The scatter of the master validation (Sec. 8, $\mathrm{RMS}_{\log} = 0.346$) sets a typical uncertainty of ±50–70% on any individual T_c^{anchor} value.

Material	T^{raw} (K)	T^{anchor} (K)	T^{lit} (K)	Comment
Class A: high-pressure superhydrides				
LuH_{10} @ 170 GPa	368	**250**	286 (DFT)	$4f^{14}$, $\mu = 0$ (flagship)
ScH_9 @ 250 GPa	350	238	180–210	$3d^1$, no $4f$
$LaBeH_8$ @ 130 GPa	287	195	185	ternary, full stability P
YbH_9 @ 550 GPa	215	146	60–100	above $P_{\mathrm{scale}}^{\mathrm{Yb}}$
$LaBeH_8$ @ 80 GPa	203	138	110 obs	kinetic limit; obs in range
LuH_6 @ 100 GPa	188	127	246	CaH_6-like, $4f^{14}$
SmH_9 @ 150 GPa	109	74	200 (no PB)	$\mu_{\mathrm{eff}} = 1.55$, moderate B_{PB}
MgH_6 @ 300 GPa	97	66	263	Mg $3s^2$
GdH_9 @ 150 GPa	**0**	**0**	∼200 (no PB)	$4f^7$ killer ($\mu = 7.94$)
Class B: ambient-pressure superhydrides				
Mg_2PtH_6	93	63	>100 with e-dop	Pt $5d^9s^1$
Mg_2IrH_6	86	58	100–160	Ir $5d^7$
Mg_2RhH_6	80	54	70–90	Rh $4d^8$
Mg_2PdH_6	76	52	80–110	Pd $4d^{10}$
Li_2AgH_6	67	45	90–120	unstable 0.319 eV/atom
Li_2AuH_6	62	42	91–140	unstable 0.172 eV/atom
Class C: bilayer nickelates (cuprate formula)				
$Ac_3Ni_2O_7$	76	114	(predicted stable)	Rhodes–Wahl, radioactive
$Ba_3Ni_2O_7$	76	113	(predicted stable)	chemical pressure
$La_3Ni_2O_7$ film	75	112	26–42 obs	film/$SrLaAlO_4$
$(La,Pr)_3Ni_2O_7$	69	103	>40 obs	15% Pr dopant
$La_2SmNi_2O_7$ @ 20 GPa	63	94	**96 obs**	agrees to 3%
$Nd_6Ni_5O_{12}$	11	16	13 obs	quintuple-layer
Class D: kagome / Class E: exotic stoichiometries				
$CsCr_3Sb_5$	**0.1**	**0.1**	1–5 (DFT)	null prediction (Cr^{3+})
ZrNH @ 150 GPa	66	66	∼40 obs (2025)	nitride-hydride
$ZrNH_2$ @ 150 GPa	54	54	15–17 obs	lower ω variant
CaC_2H_8 @ 30 GPa	78	78	80–100 (DFT)	p-block clathrate

Table 6: TGP predictions for currently investigated superconductor candidates. All TGP parameters are frozen at the values established in Sections 3–7; no per-candidate refit is performed. Material inputs $(a, \omega, z, \eta, \mathrm{orb}, \mu_{\mathrm{eff}})$ are taken from published DFT structures and atomic Hund tables. Full script: `ps22_open_candidates_predictions.py`.

Three structural features of Table 6 are worth emphasising.

- *Null predictions with teeth.* TGP predicts $T_c \approx 0$ K for GdH_9 (and by extension TbH_9, DyH_9, HoH_9, ErH_9, TmH_9), even though naive Eliashberg estimates without pair-breaking give ~ 200 K for each. A single experimental confirmation that, e.g., synthesised GdH_9 stays normal above 1 K decisively supports Eq. (12); a confirmed $T_c \gtrsim 100$ K falsifies it. Likewise, $CsCr_3Sb_5$ is forbidden as a conventional s-wave superconductor by TGP; any $T_c \gtrsim 10$ K forces an unconventional channel (flat-band, topological) outside the present framework.

- *Quantitative disagreements with naive DFT.* For ambient Li_2AuH_6 and Mg_2PtH_6, Eliashberg calculations in the literature give $T_c = 91$–140 K, whereas frozen-parameter TGP gives 42–63 K

— a factor-two discrepancy rooted in $M(a)$ and in the reduced $\eta(P=0)$ for p-block-centred cages. If any such compound is eventually stabilised at ambient pressure, the measured T_c will directly discriminate between the two frameworks.

- *Class C internal tension.* Applying the cuprate P6.A formula to bilayer nickelates gives $T_c^{\text{anchor}} \sim 100\,\text{K}$ for thin-film $La_3Ni_2O_7$, whereas the observed onset is $\sim 40\,\text{K}$. One resolution is to introduce a nickelate-specific $\Lambda_E^{\text{Ni}} \approx \Lambda_E^{\text{cup}}/3$, consistent with the weaker J_{AF} of Ni d^9 versus Cu d^9. The fact that $La_2SmNi_2O_7$ at 20 GPa agrees to 3% without any such rescaling suggests that the "effective Λ_E" is pressure and doping dependent — a structural prediction that can be tested by a single-crystal doping-vs-T_c scan.

The table demonstrates that TGP generates a broad spectrum of *parameter-free* predictions — including several that are qualitatively at odds with the current consensus (the $4f^7$ killer series, the ambient-superhydride tower, the nickelate Λ_E rescaling). Each constitutes an independent falsification channel that is within experimental reach on a few-year horizon.

10 Discussion and limitations

The framework presented here reduces the specification of T_c across five superconductivity classes to three universal TGP constants plus a small set of tabulated atomic inputs. The reduction is most striking for magnetic blocking: eleven phenomenological λ_{sf} values, once material-specific, become a single universal coupling $\kappa_{\text{TGP}} = 2.012$ plus $\{N(E_F), I\}$. And the introduction of a Hund moment-dependent pair-breaker extends the framework to the lanthanide hydride class without any class-specific refit.

Systematic limitations.

(L1) The cuprate layer factor $\sqrt{n}$ is an approximation; residuals suggest $n^{0.7}$ with an offset. Open.

(L2) Ce is only partially captured: Kondo-screened μ_{eff} reduces the naive pair-breaker. A unified hybridisation treatment (P7.3) is needed.

(L3) AFM nesting ($\mathbf{q} \neq 0$ susceptibility, e.g. Cr) is outside the paramagnon-mediated substrate treatment.

(L4) Ordered ferromagnets ($IN \geq 1.5$: Co, Ni) are over-predicted for λ_{sf} by factor 2–3, but all such cases are already magnetically suppressed to $T_c \to 0$, so this has no observational consequence.

(L5) Eu and Yb valence transitions under pressure ($Eu^{2+} \to Eu^{3+}$ at $\sim$20 GPa; Yb in metal as Yb^{2+}) require a P-dependent μ_{eff} table. Open.

Ambient-pressure room-temperature SC. The framework admits no known ambient, room-temperature, stoichiometric candidate. The most accessible path is Hg1245 + pressure-quench (extending the Hg1223-quench at 151 K reported by [8]) or BC_3/BC_4 synthesis. We consider ambient RT-SC open; the present framework does not preclude it, but also does not predict a specific candidate material among those currently synthesised.

Cross-sector consistency with other TGP applications. The substrate inputs used here — the harmonic $a^*_{\text{TGP}} = 4.088\,\text{Å}$ and the orbital-amplitude spectrum $\{A_s, A_{sp}, A_d, A_f\}$ — are *not* superconductivity-specific: they are fixed by the core theory [1] and reused without rescaling in several independent applications that are developed in companion analyses. In particular, the universal substrate constant $c_{\text{TGP}} = 1/(4\pi)$ appears (i) as the leading-order coefficient in the TGP derivation of the Kovtun–Son–Starinets viscosity bound $\eta/s \geq 1/(4\pi)$ for liquids (see `research/liquid_viscosity/`); (ii) as the prefactor in the substrate self-interference potential

that fixes the cosmological $\Lambda_E = \gamma/12$ limit [1]; and (iii) as the normalisation of the Casimir-force correction in MOFs (`research/casimir_mof/`). The three superconductivity-specific constants isolated here — $\beta = 2.527$, $\kappa_{\rm TGP} = 2.012$, $\alpha_{\rm PB} = 0.2887\,\mu_B^{-2}$ — are channels-specific couplings within the universal substrate framework, not independent fitting parameters: each is fixed once on its calibration subset and then used unchanged across the remainder of the 31-material dataset.

11 Conclusion

Building on the TGP framework of [1], we have developed a unified model of superconductivity that predicts T_c across five classes (cuprates, phonon-mediated, pressure-driven, iron-based, and lanthanide hydrides) using a single substrate ansatz modified by three physical blockers. Three universal TGP constants — $\beta = 2.527$, $\kappa_{\rm TGP} = 2.012$, $\alpha_{\rm PB} = 0.2887\,\mu_B^{-2}$ — plus tabulated atomic inputs yield $r(\log T_c) = 0.875$ across thirty-one materials without class-specific refit. The reduction of eleven empirical $\lambda_{\rm sf}$ values to a single TGP-structured closed-form expression (Eq. (10)), with one fitted coupling $\kappa_{\rm TGP}$ calibrated on five d-metals, *improves* the Pearson correlation from 0.874 to 0.943 while reducing the parameter count from eleven to one; this is the strongest internal validation of the TGP substrate picture in the superconducting sector.

Five falsifiable predictions are stated; the single most distinctive is the LuH_{10} high-pressure prediction of 200–250 K, arising from the $4f^{14}$ closed-shell structure of Lu^{3+} and the vanishing Hund moment.

Data and reproducibility. All numerical scripts (`ps10`–`ps21`), master validation (`ps20`), and summary documents (`P6A-D`, `P7A-B`) are deposited in the companion repository [2].

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